macroclimate

ibdj

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1 Macroclimate data sources

Background macroclimate of Kangerluarsunnguaq (Kobbefjord, Nuuk 64.13334, -51.34368). Data produced by Asiaq - Greenland Survey for Greenland Ecosystem Monitoring (GEM 2020c, 2020a, 2020b) and DMI (Cappelen 2021; Jensen 2024).

2 Temperature data

The imported data has entries between 2007 and 2022.

Temperatures between -30.2°C (2008-02-11) and 23.2°C (2016-06-10) have been recorded.

2.1 Mean annual temperature

The mean temperature across all entries is -0.1003°C.

The warmest months are July (10.49°C), August (9.49°C) and June (7.25°C).

The coldest months are February (-8.97°C), March (-7.97°C), and January (-7.34°C).

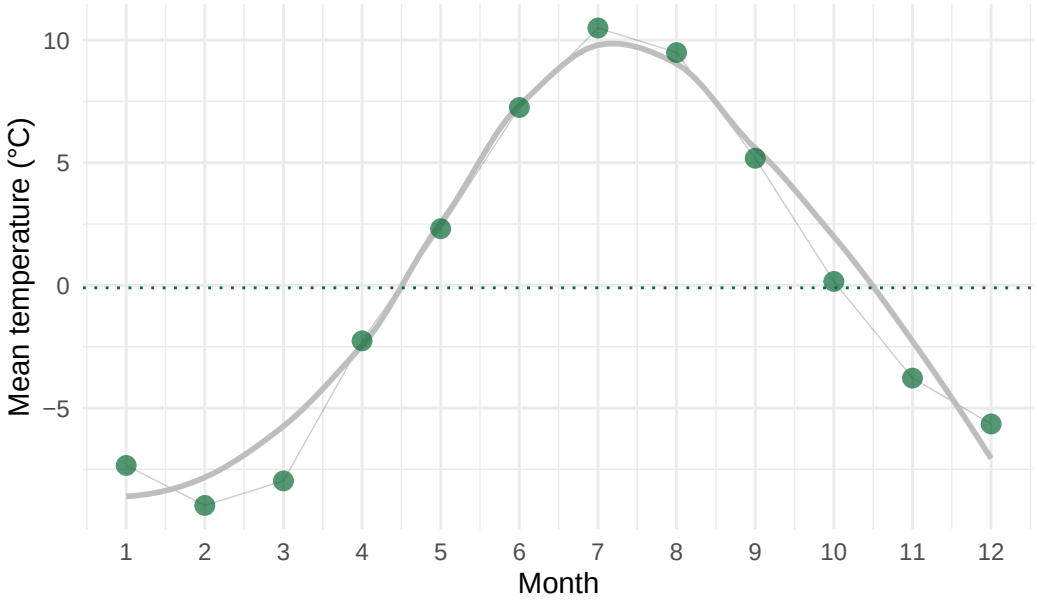

Figure 1: Mean temperature pr month. Mean air temperature (°C) pr month, based on data from 2007 to 2022. The dashed line represent yearly mean of -0.1°C.

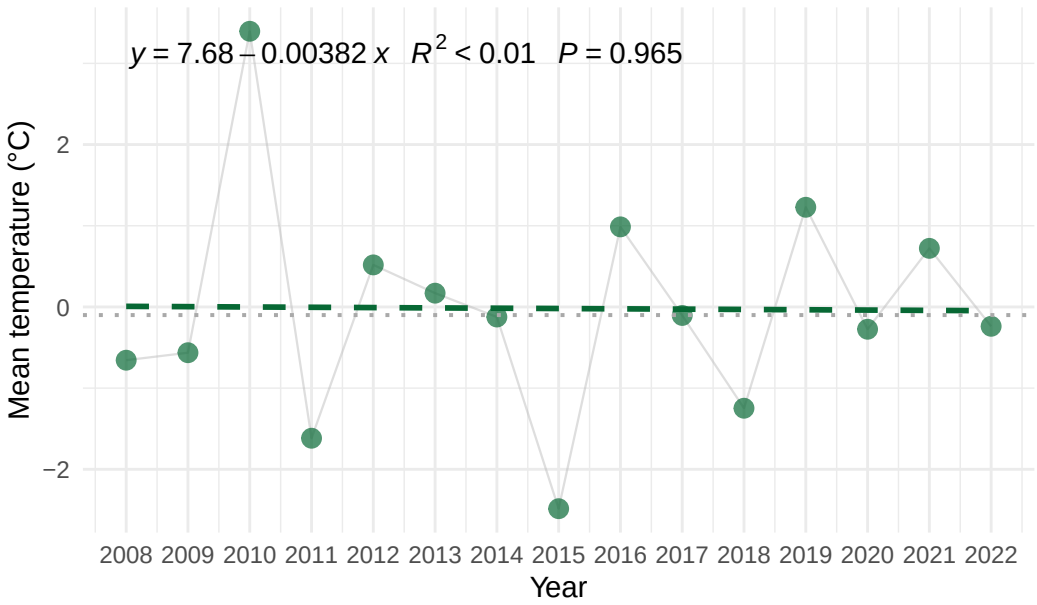

Figure 2: Annual mean temperature, Kangerluassunguaq. Mean air temperature pr year (°C) from 2008 to 2022. Data from 2007 is excluded because data is only from October, November and December. Gray dotted line indicate over all mean of -0.1003°C. Dashed line is trend line.

2.2 Mean temperature of warmest months

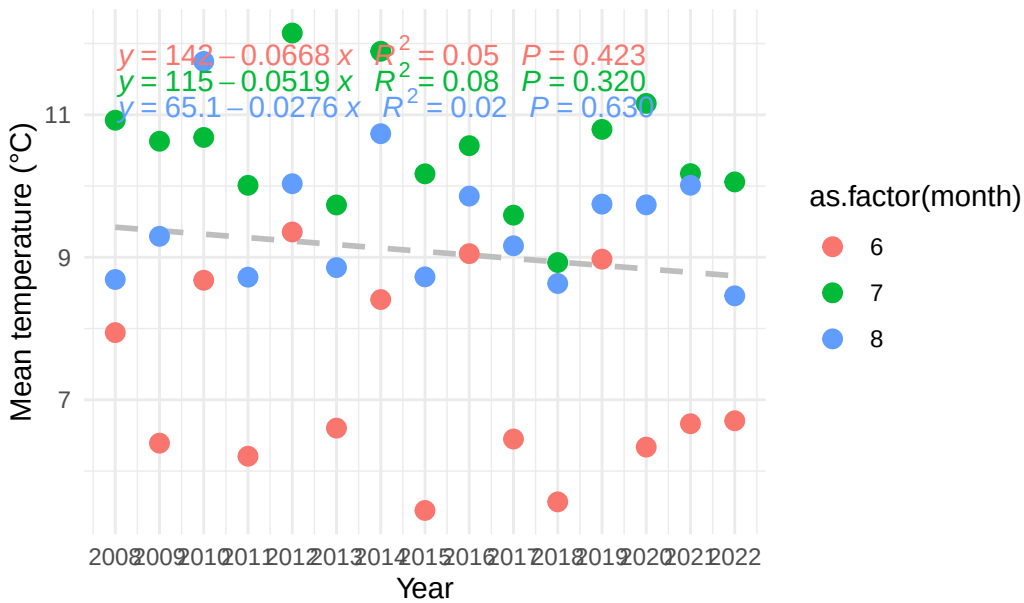

Figure 3: Mean temperature pr summer month. Mean air temperature (°C) pr for July, August, and June based on data from 2007 to 2022. P-value for the regression i 0.416. The mean of the warmests months across all years are 9.081°C.

```
Call:
lm(formula = mean ~ year, data = data_temp_monthly_mean_summer)

Residuals:
    Min      1Q  Median      3Q     Max
-3.6340 -0.7226  0.1279  1.2551  2.9202

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 107.34623  119.65318   0.897    0.375
year         -0.04877    0.05938  -0.821    0.416

Residual standard error: 1.721 on 43 degrees of freedom
Multiple R-squared:  0.01544,   Adjusted R-squared:  -0.007454
F-statistic: 0.6745 on 1 and 43 DF,  p-value: 0.416
```

2.3 Temperature anomalies

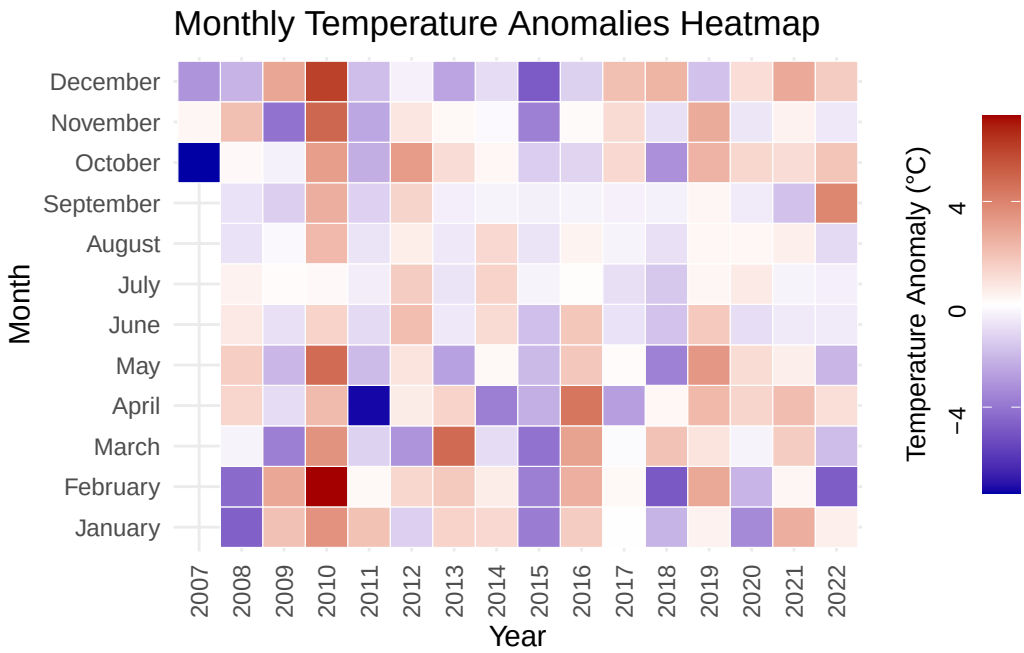

Figure 4: Monthly Temperature Anomalies Heatmap. The stronger the color the bigger the difference from mean temperature of the respective month across all monitored years (°C).

2.4 Absolute values of anomolies

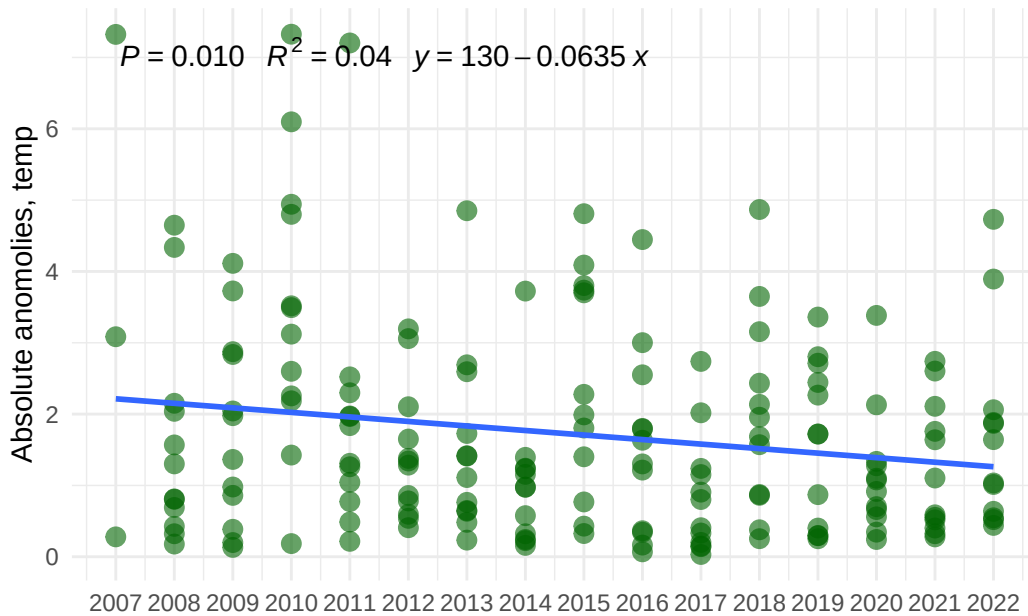

Figure 5: Absolut values of anomolies (difference from mean) of respective month.

2.5 5 year intervals temperature

```
 segment    emmean    SE  df asymp.LCL asymp.UCL .group
 2022-2026 -0.2385 1.530 Inf     -3.24      2.76  1
 2012-2016 -0.1872 0.684 Inf     -1.53      1.15  1
 2017-2021  0.0643 0.684 Inf     -1.28      1.41  1
 2007-2011  0.1395 0.765 Inf     -1.36      1.64  1

Degrees-of-freedom method: asymptotic
Confidence level used: 0.95
P value adjustment: tukey method for comparing a family of 4 estimates
significance level used: alpha = 0.05
NOTE: If two or more means share the same grouping symbol,
      then we cannot show them to be different.
      But we also did not show them to be the same.
```

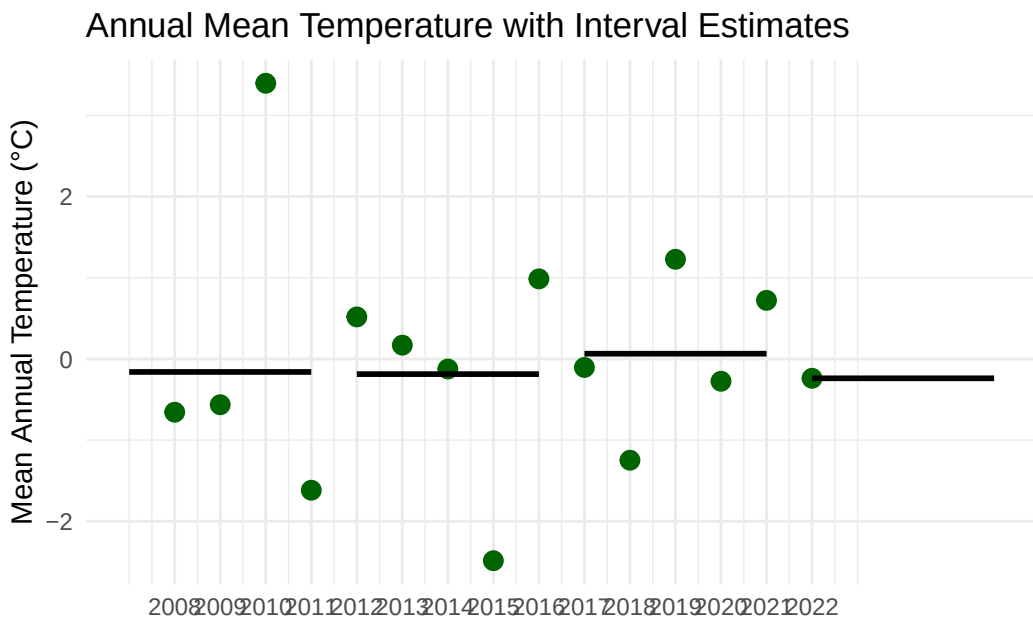

Figure 6: 5 year intervals in temperature means.

3 Precipitation data

3.1 Yearly precipitation

The mean annual precipitation is 886.15 mm. 874.0611111

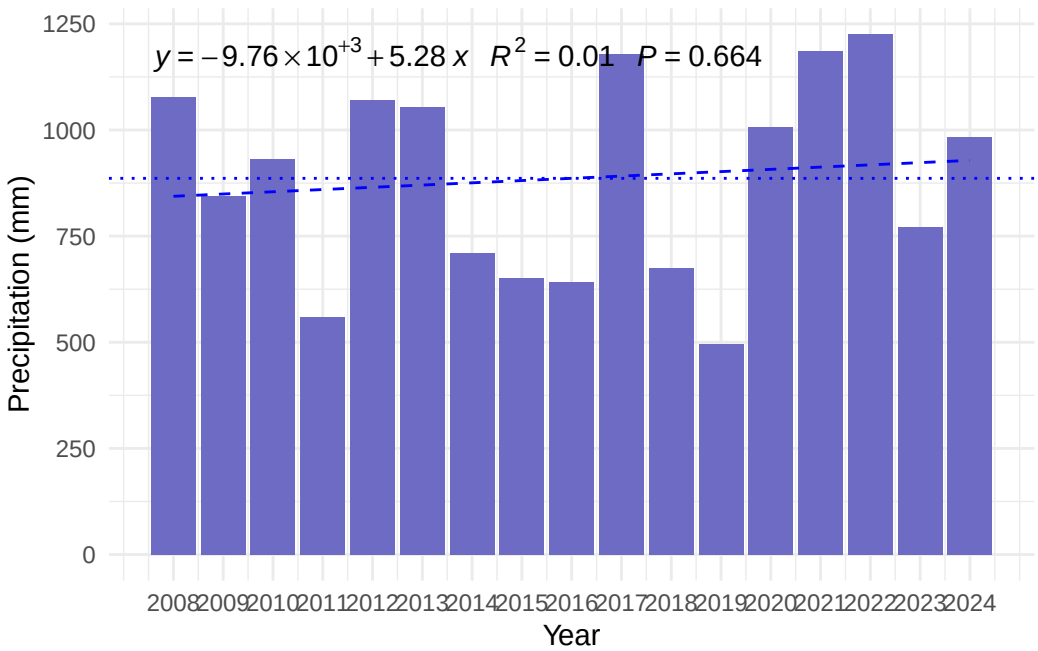

Figure 7: Mean yearly precipitation (mm) from 2008 to 2024. Data from 2007 is excluded because it was only from May - December. Blue dashed line indicate over all mean of 886.15 mm.

3.2 Monthly precipitation

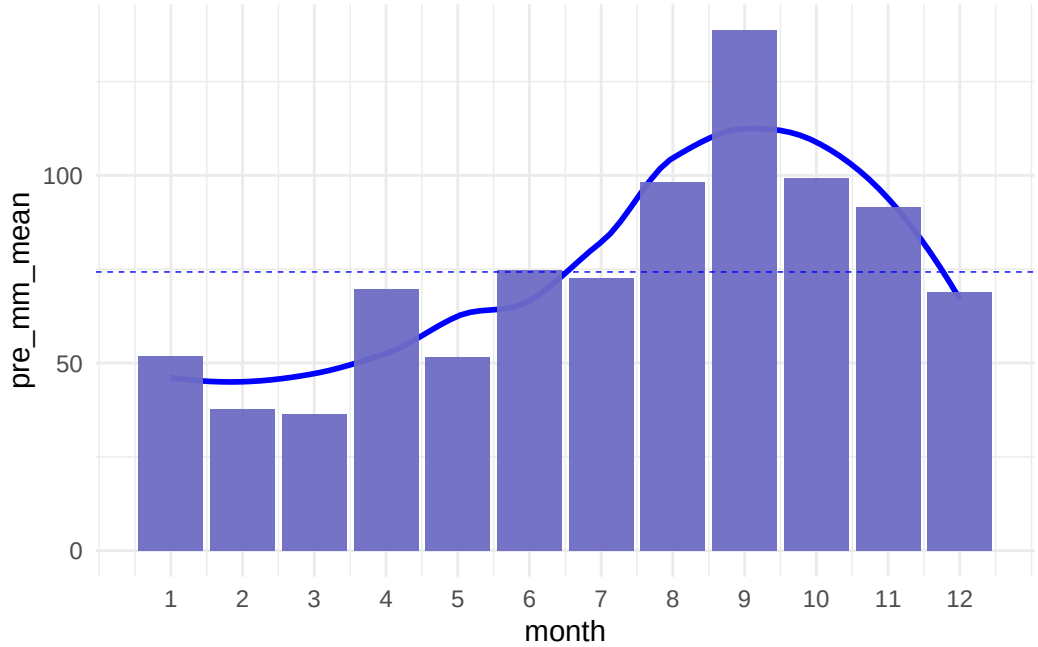

Figure 8: Mean monthly precipitation (mm) including data from 2007 to 2024. Blue dashed line represent monthly mean of 74.27 mm. ADD SF BARS!

The months with the most precipitation are September (138.64 mm), October (99.38 mm), August (98.28 mm), which account for 336.3 mm or 37.95 % of the yearly precipitation.

The months with the least precipitation are May (51.51 mm), February (37.62 mm), March (36.44 mm).

3.3 Precipitation type

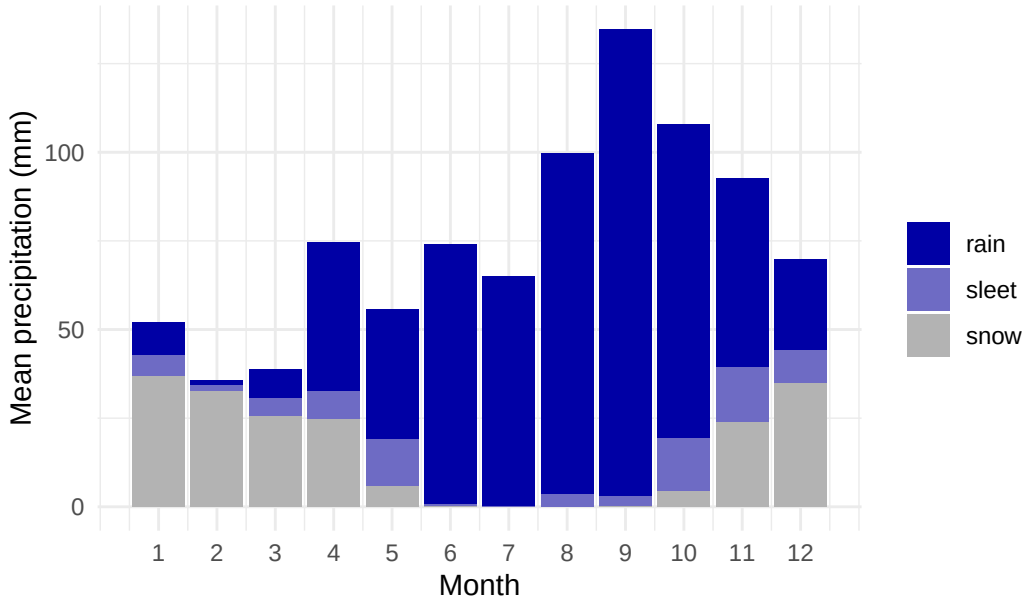

Figure 9: Mean precipitation type pr month. Mean monthly distribution of types of precipitation (mm). Based on data from 2008 to 2022. Sleet is defined as precipitation that fell when temperatures were between -1°C and 1°C.

Out of 886.15 (900.8) mm mean annual precipitation, rain accounts for 631.55 mm (70.1 %), snow accounts for 188.71 mm ((20.9 %)), and sleet for the remaining 80.54 mm (8.9 %).

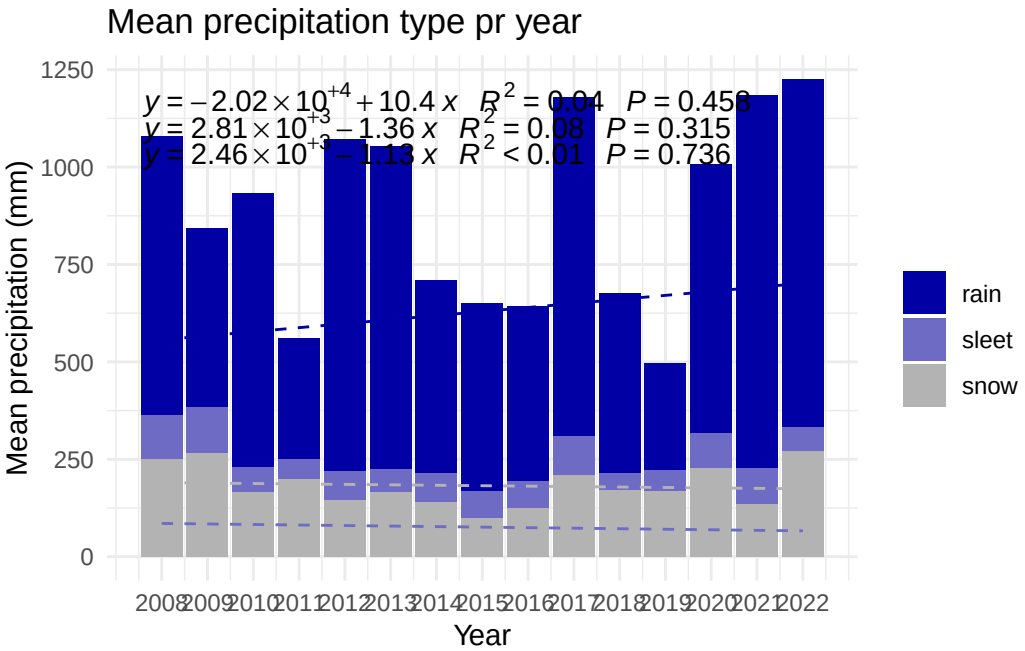

Figure 10: Mean percipitation pr. month over the monitoring period (2008 - 2022). Sleet is defined as precipitation that fell when temperatures were between -1°C and 1°C. (Data only to 2022 because the temperature data is only till 2022.)

3.4 Anomalies

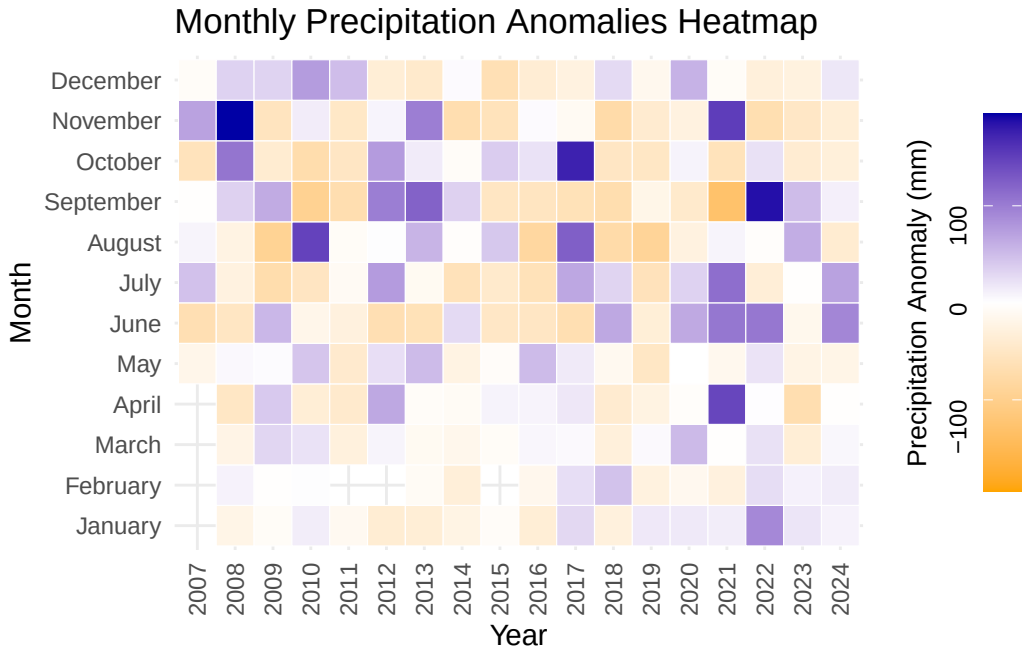

Figure 11: Monthly precipitation anomaly (mm). Stronger colors indicate large derivation from the month mean. Anomalies vary from -130.04 to 194.12.

3.5 Absolute values of anomolies

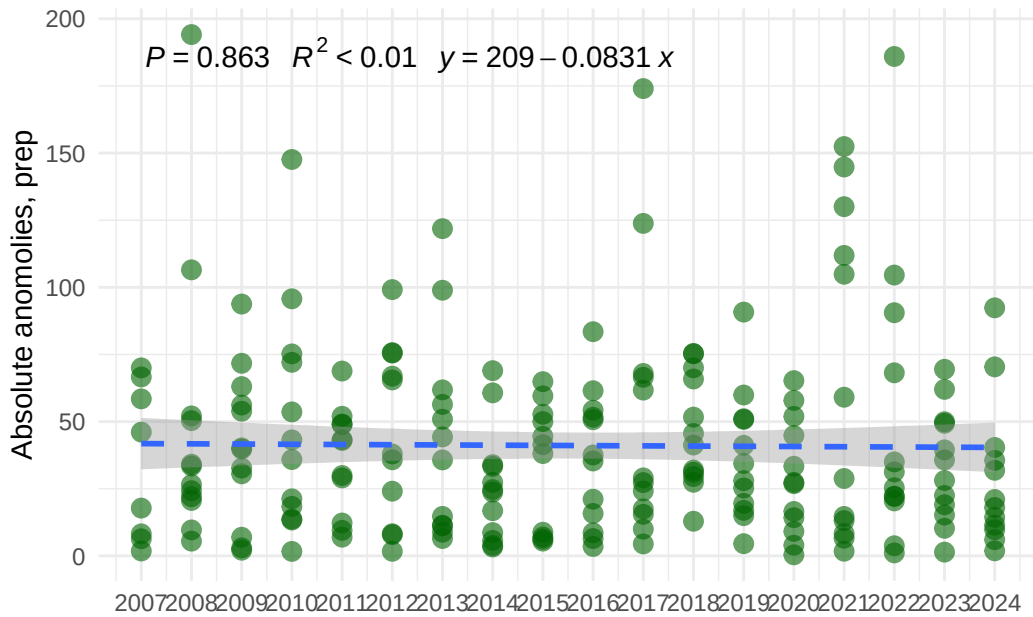

Figure 12: Absolut values of anomolies (difference from mean) of respective month.

3.6 5 year intervals precipitation

```
     date                 time                      pre_mm
 Min.   :2008-01-01   Min.   :00:00:00.000000   Min.   : 0.0000
 1st Qu.:2012-07-23   1st Qu.:06:00:00.000000   1st Qu.: 0.0000
 Median :2016-09-30   Median :12:00:00.000000   Median : 0.0000
 Mean   :2016-08-30   Mean   :11:30:21.122933   Mean   : 0.1058
 3rd Qu.:2020-12-09   3rd Qu.:18:00:00.000000   3rd Qu.: 0.0000
 Max.   :2024-12-31   Max.   :23:00:00.000000   Max.   :19.0000
 quality_flag            year          month              doy
 Length:142395      Min.   :2008   Min.   : 1.000   Min.   :  1.0
 Class :character   1st Qu.:2012   1st Qu.: 4.000   1st Qu.: 99.0
 Mode  :character   Median :2016   Median : 7.000   Median :187.0
                    Mean   :2016   Mean   : 6.625   Mean   :186.3
                    3rd Qu.:2020   3rd Qu.:10.000   3rd Qu.:276.0
                    Max.   :2024   Max.   :12.000   Max.   :366.0
 month_string           anchor       seg_start       seg_end
 Length:142395      Min.   :2007   Min.   :2007   Min.   :2011
 Class :character   1st Qu.:2007   1st Qu.:2012   1st Qu.:2016
 Mode  :character   Median :2007   Median :2012   Median :2016
```

```
                    Mean   :2007   Mean   :2014   Mean   :2018
                    3rd Qu.:2007   3rd Qu.:2017   3rd Qu.:2021
                    Max.   :2007   Max.   :2022   Max.   :2026
  segment           seg_mean_precip
 Length:142395      Min.   :2980
 Class :character   1st Qu.:3413
 Mode  :character   Median :4128
                    Mean   :3877
                    3rd Qu.:4544
                    Max.   :4544

 segment   emmean  SE df lower.CL upper.CL .group
 2012-2016    826 112 13      583     1068  1
 2007-2011    853 126 13      582     1125  1
 2017-2021    909 112 13      666     1152  1
 2022-2026    993 145 13      680     1307  1

Confidence level used: 0.95
P value adjustment: tukey method for comparing a family of 4 estimates
significance level used: alpha = 0.05
NOTE: If two or more means share the same grouping symbol,
      then we cannot show them to be different.
      But we also did not show them to be the same.

 segment    emmean    SE  df asymp.LCL asymp.UCL .group
 2022-2026 -0.2385 1.530 Inf     -3.24      2.76  1
 2012-2016 -0.1872 0.684 Inf     -1.53      1.15  1
 2017-2021  0.0643 0.684 Inf     -1.28      1.41  1
 2007-2011  0.1395 0.765 Inf     -1.36      1.64  1

Degrees-of-freedom method: asymptotic
Confidence level used: 0.95
P value adjustment: tukey method for comparing a family of 4 estimates
significance level used: alpha = 0.05
NOTE: If two or more means share the same grouping symbol,
      then we cannot show them to be different.
      But we also did not show them to be the same.
```

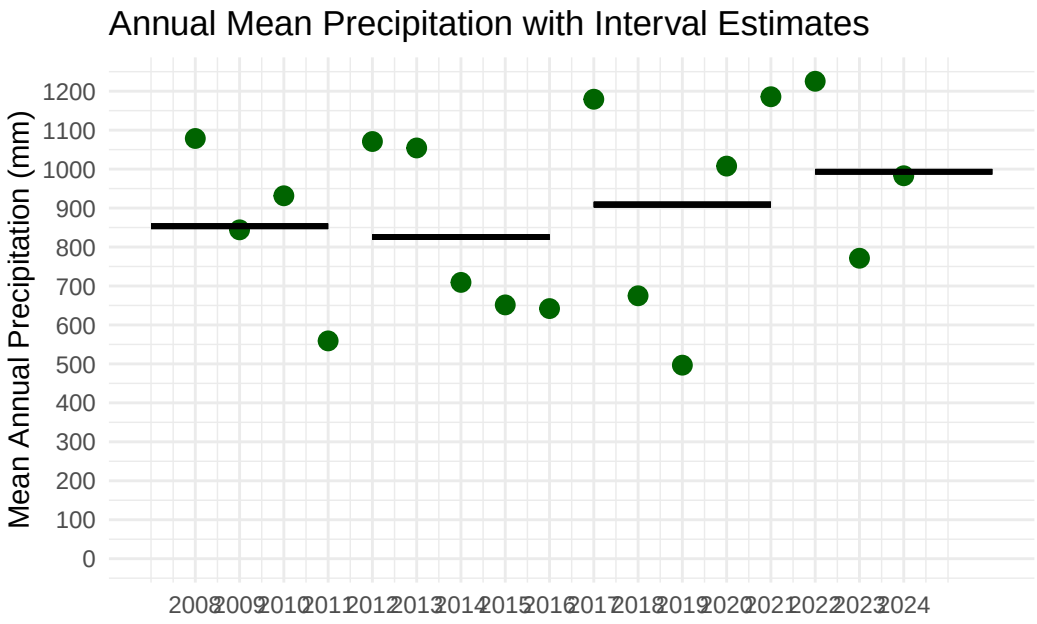

Figure 13: 5 year intervals in precipitation means.

4 PAR data

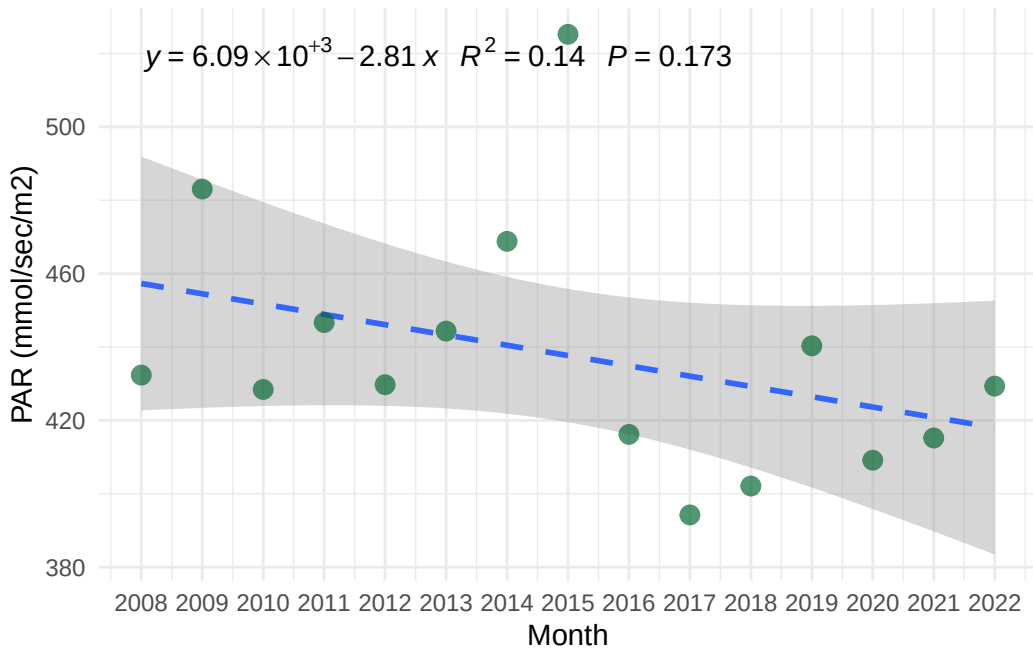

Figure 14: Yearly mean PAR from 2008 to 2022. Calculations based on PAR values only when the sun is up (solar elevation > 0).

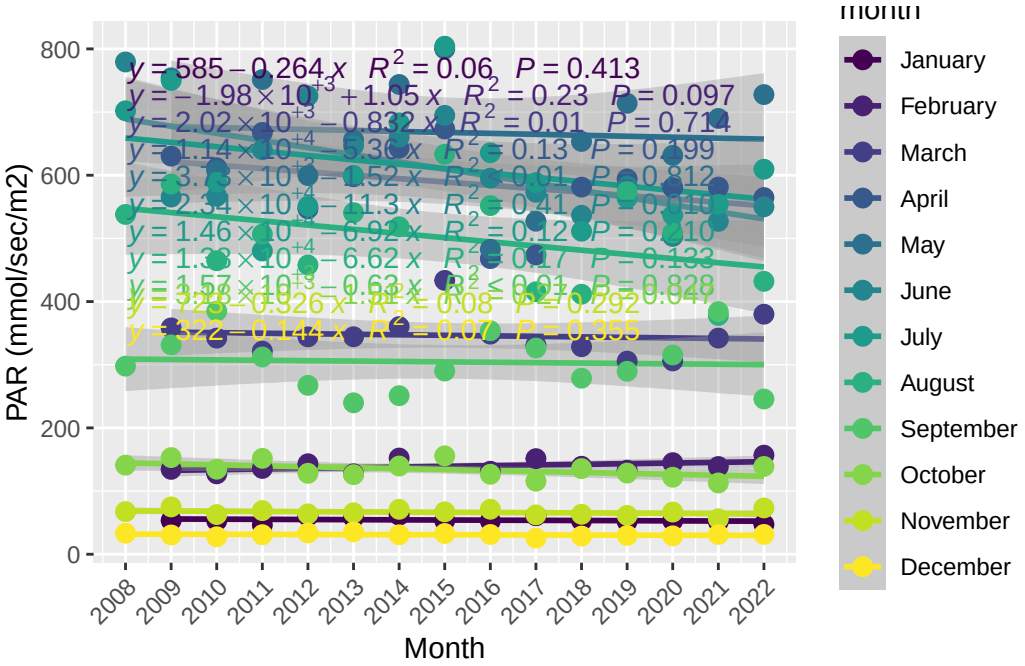

Figure 15: Monthly mean PAR from 2008 to 2022. Calculations based on PAR values only when the sun is up (solar elevation > 0).

4.1 Monthly means

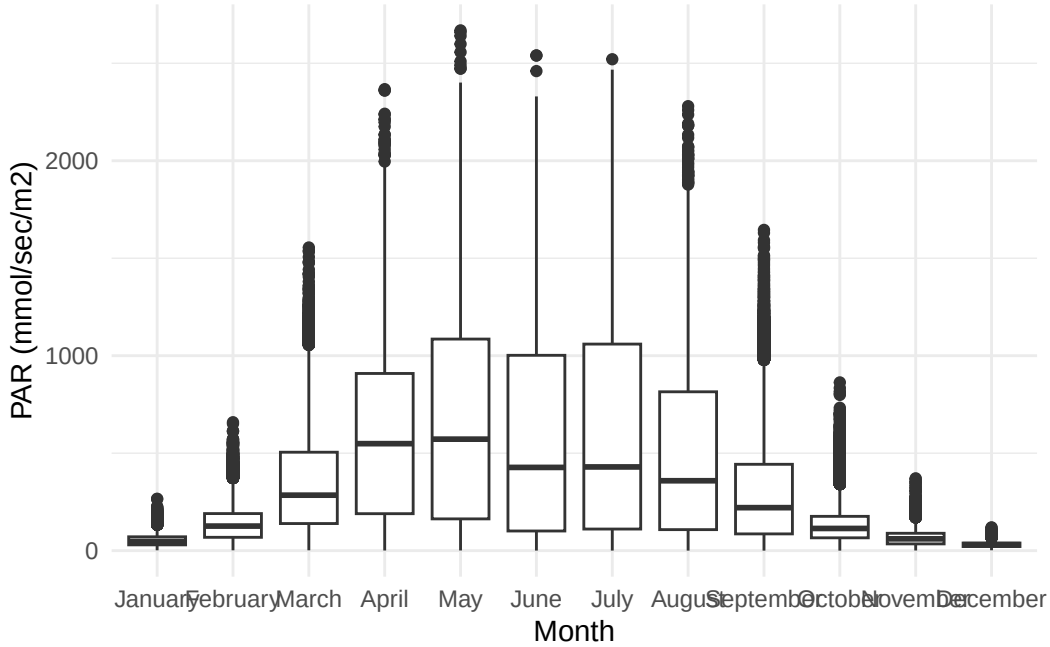

Figure 16: Monthly mean PAR from 2008 to 2022. Calculations based on PAR values only when the sun is up (solar elevation > 0).

4.2 Highest monthly means

The months with the highest mean levels of PAR (when the sun is up, solar elevation > 0) are May (663.1), July (610.3), June (597).

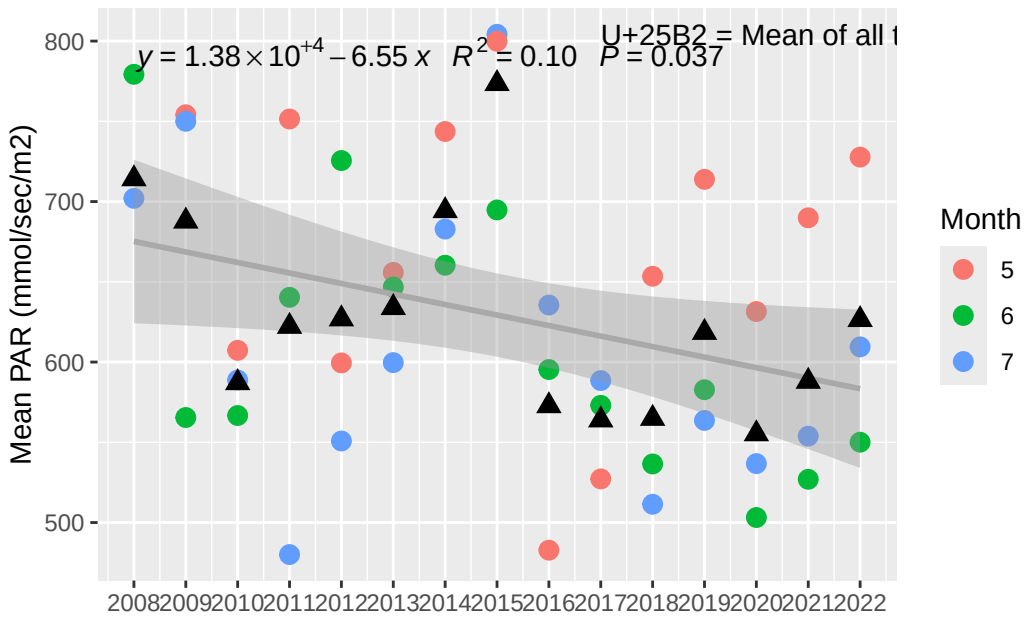

Figure 17: Mean of months with highest PAR.

Data indicate that high-radiation months have gotten less radiant during the monitoring period (see Figure 17).

```
Call:
lm(formula = par_mean ~ year, data = par_highest_mean)

Residuals:
    Min      1Q  Median      3Q     Max
-175.44  -60.59  -17.74   68.27  174.95

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 13835.416   6118.291   2.261   0.0290 *
year           -6.554      3.036  -2.159   0.0366 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 85.33 on 42 degrees of freedom
Multiple R-squared:  0.09987,   Adjusted R-squared:  0.07843
F-statistic:  4.66 on 1 and 42 DF,  p-value: 0.03664
```

4.3 Mean pr month

4.4 Observations above 2000

```
Call:
lm(formula = above_2000 ~ year, data = above_2000)

Residuals:
   Min     1Q Median     3Q    Max 
-8.299 -6.156 -3.513  4.737 27.486 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 152.8806   646.6144   0.236    0.814
year         -0.0714     0.3210  -0.222    0.825

Residual standard error: 8.88 on 41 degrees of freedom
Multiple R-squared:  0.001205,	Adjusted R-squared:  -0.02316 
F-statistic: 0.04946 on 1 and 41 DF,  p-value: 0.8251
```

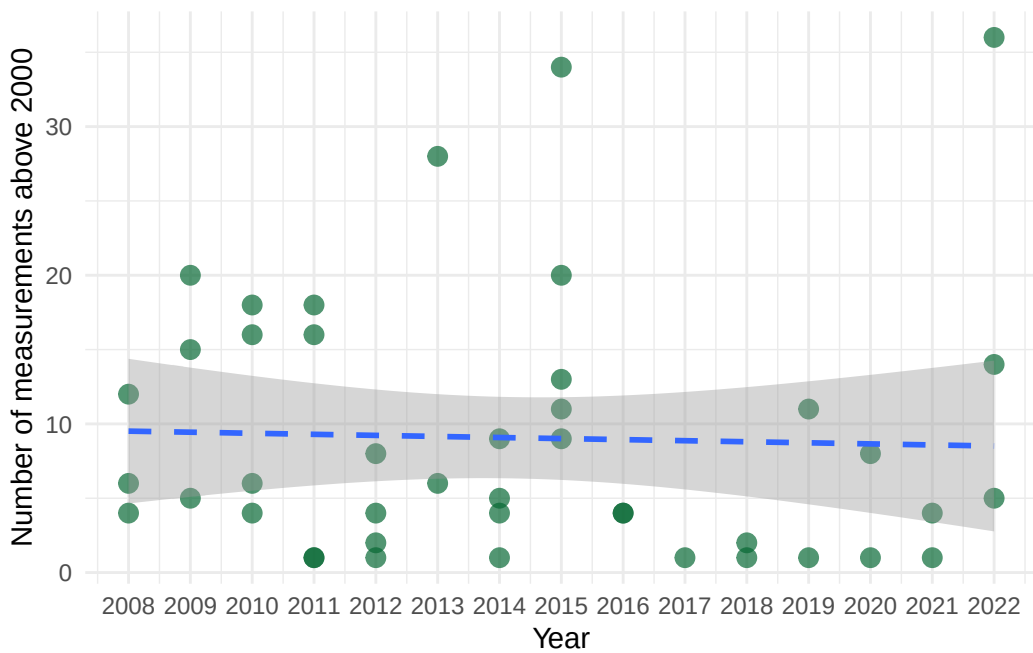

Figure 18: Number of measurements of PAR above 2000.

4.5 Observations below 200

```
Call:
lm(formula = below_200_count ~ year, data = data_par_below_200)

Residuals:
    Min      1Q  Median      3Q     Max 
-30.450  -4.668   2.586   6.332  13.971 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -2294.807   1376.234  -1.667    0.119
year            1.193      0.683   1.747    0.104

Residual standard error: 11.43 on 13 degrees of freedom
Multiple R-squared:   0.19, Adjusted R-squared:  0.1277 
F-statistic:  3.05 on 1 and 13 DF,  p-value: 0.1043
```

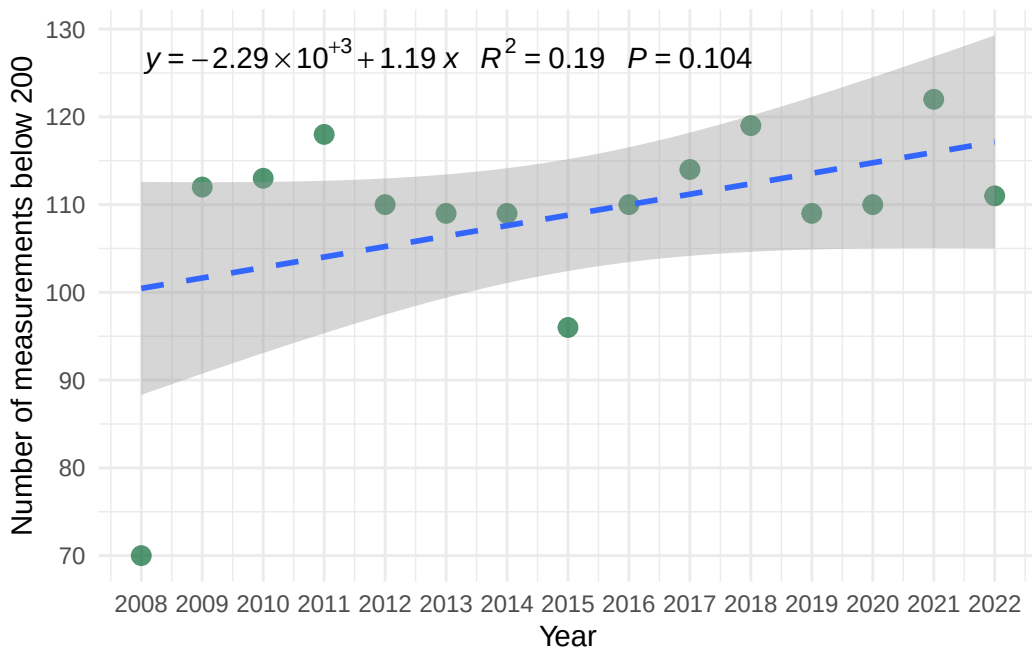

Figure 19: Number of days pr year with mean par below 200.

4.6 Anomalies

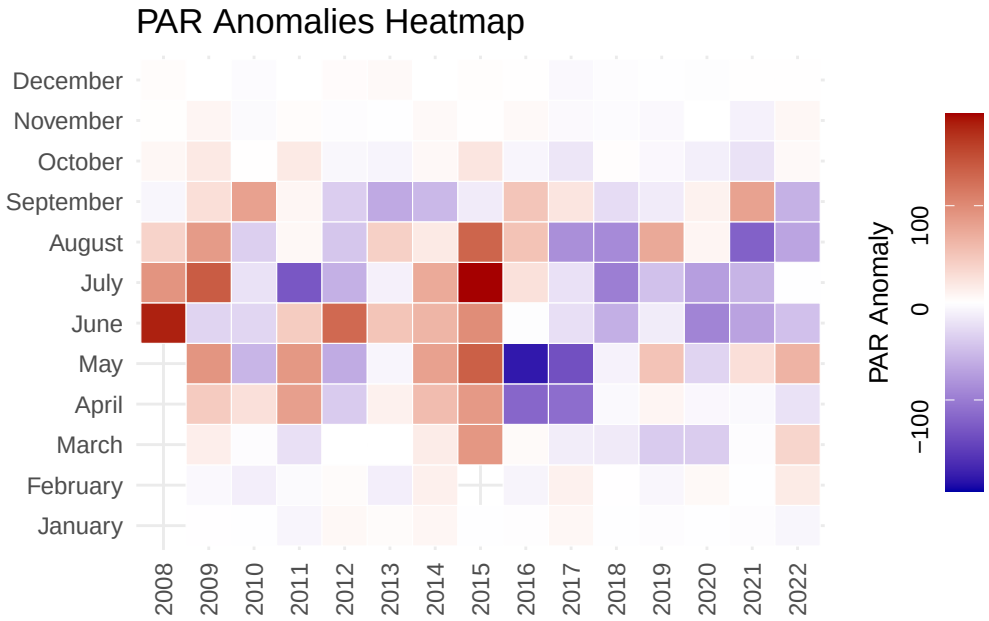

Figure 20: Anomolies (difference from mean) of respective month. Stronger color indication bigger difference from monthly mean across entire period. Strongest color corresponding to anomaly of -180.3 (darkest blue) - 193.9 (darkest red).

4.7 Absolute values of anomolies

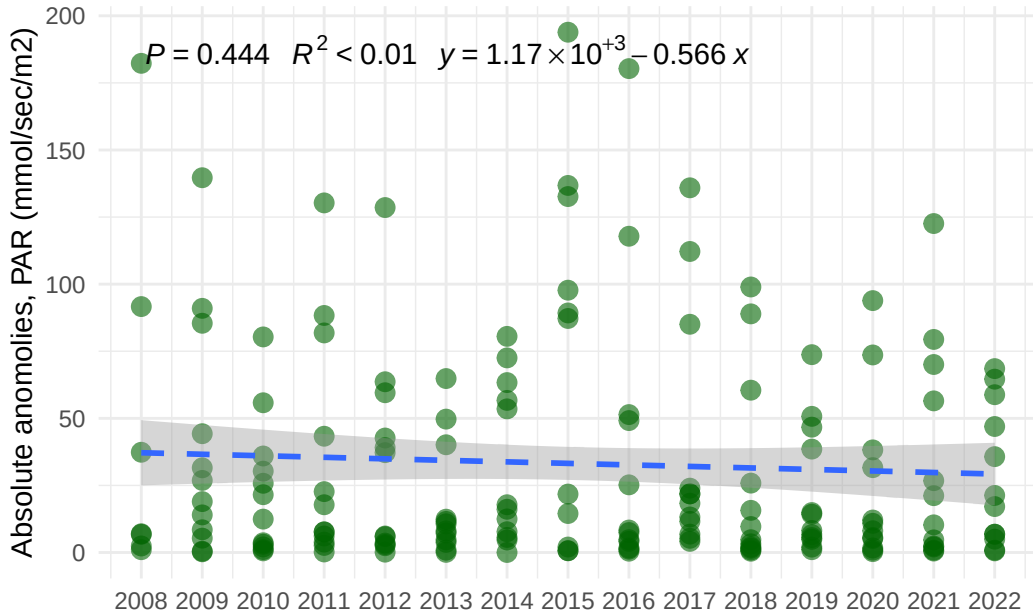

Figure 21: Absolut values of anomolies (difference from mean) of respective month.

5 Degree days

The number of degree days (days with a mean temperature > 0°C) has not changed significantly during the monitoring period.

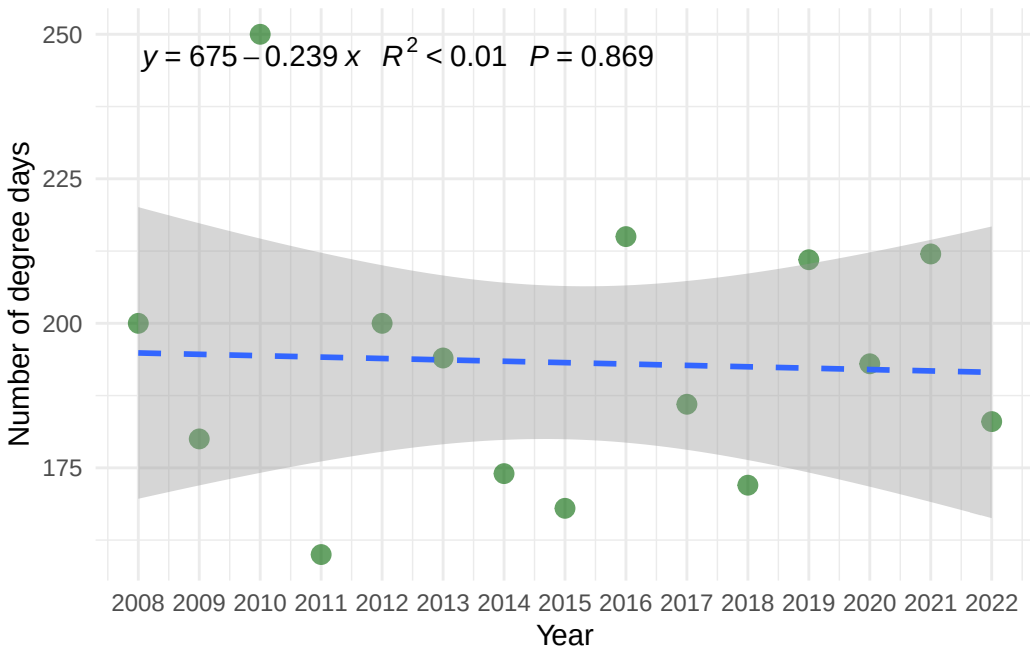

Figure 22: Degree days, days with mean temperature above 0°C

5.1 Degree days anomilies (in number of days)

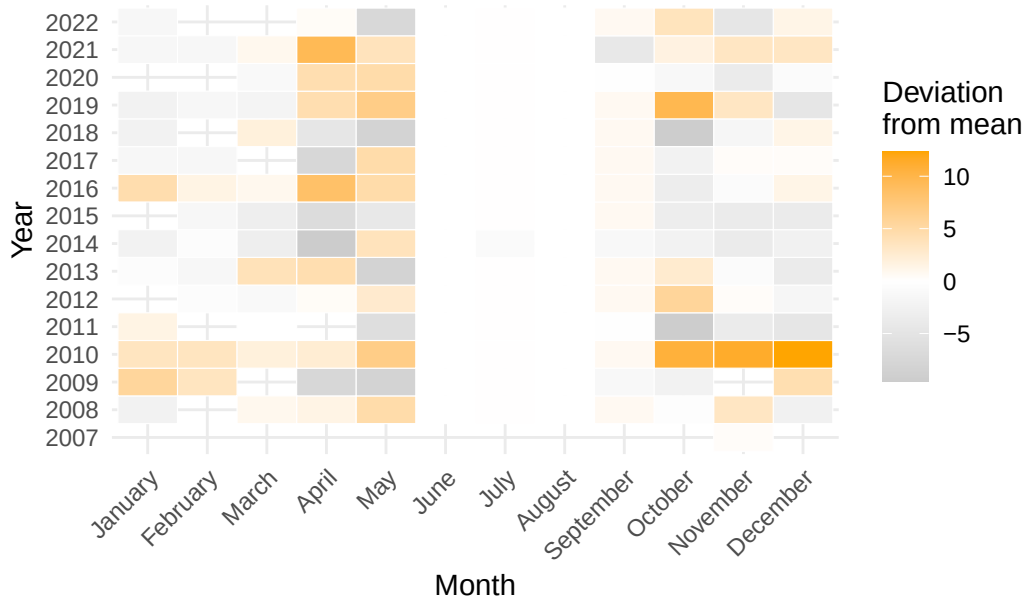

Figure 23: Degree Day Anomalies by Month-Year. Degree days, days with mean temperature above 0°C

5.2 Absolute numbers of variation in degree days

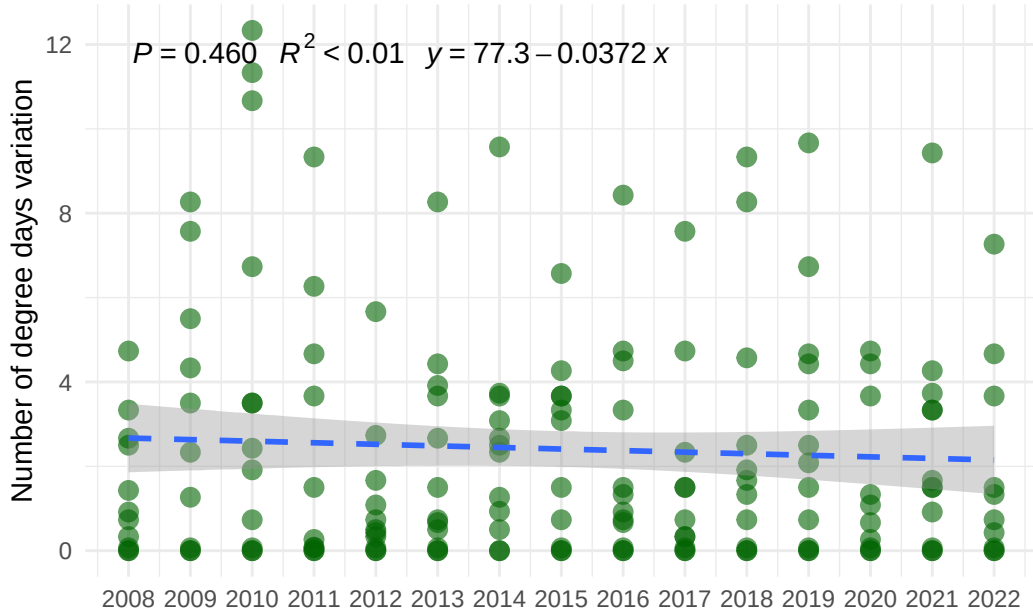

Figure 24: Absolute numers of deviation from mean number of degree days

6 Historic climate context

6.1 CHELSA

Lookings to see if CHELSA data shoc the same pattern of minimal change as the local data. Using this guide to optain data: https://gitlabext.wsl.ch/karger/rchelsa

6.2 DMI

Looking at the historic weather data from DMI from Nuuk. I.e. not the exact location of Kangerluarsunnguaq, but Nuuk. Comparing overall trends with data from (Cappelen 2021; Jensen 2024).

- For rain
- For temp
- Only for year or monthly trends too?
- Station Nuuk is (3)4250

- 601 is accumulated precipitation
- for 25-08: 101 is Mean air temperature (°C; 2 meters above ground).

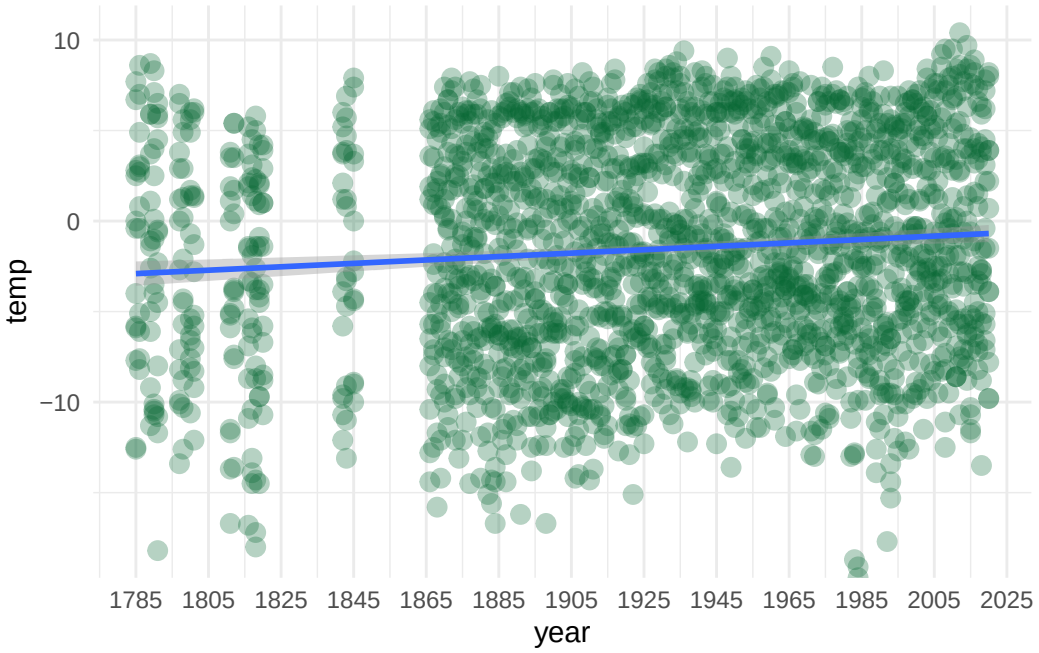

Figure 25: Mean air temperature (°C) pr year, from 1784 to 2020. Yearly mean of 173.6°C. Yearly mean ranges from -46.1°C to 1785°C.

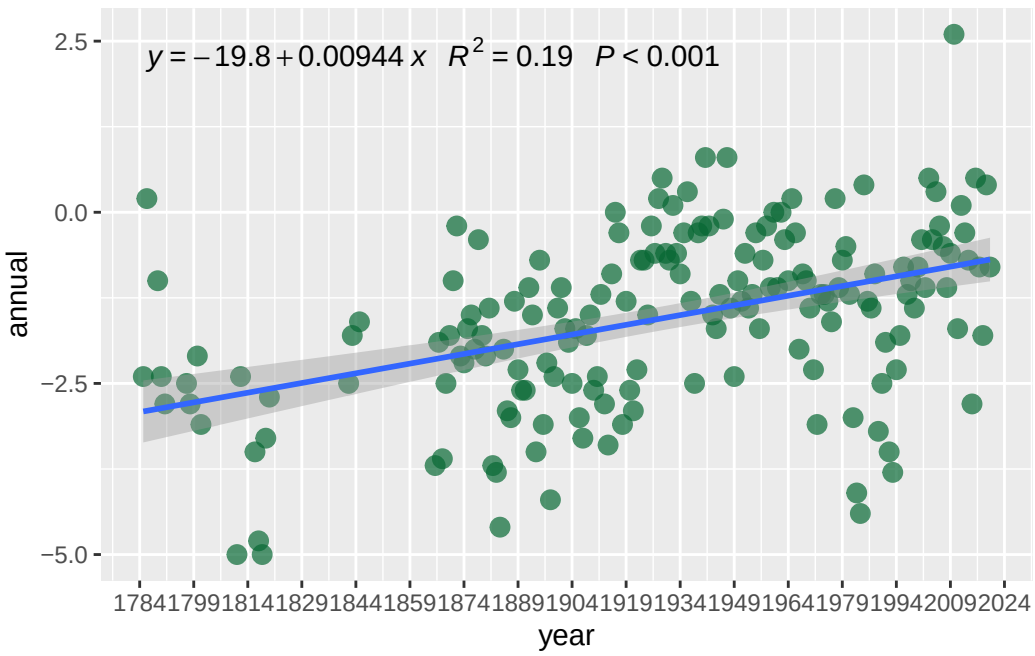

Figure 26: Mean air temperature (°C) pr year, from 1785 to 2020. Yearly mean of -1.6°C. Yearly mean ranges from -5°C to 2.6°C. Mean temperature has increase from -2.448°C in 1784 to 1.272°C in 2024

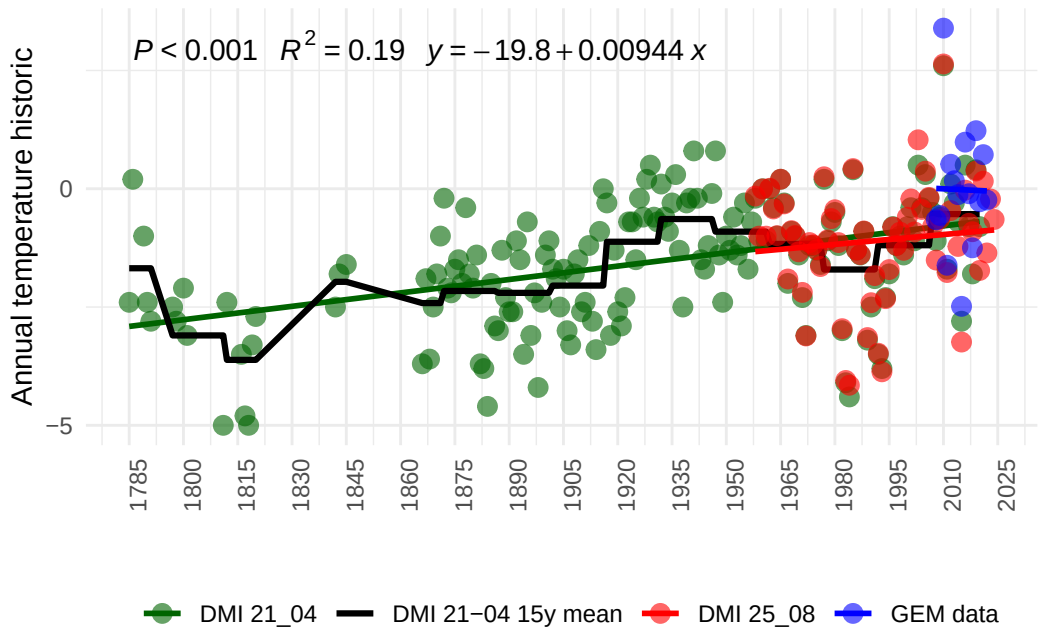

Figure 27: Annual historic temperature in °C comparing data from DMI 21-04, DMI 25-08 and GEM. Black line represent 15 year interval means for DMI 21-04.

```
tibble [16 x 2] (S3: tbl_df/tbl/data.frame)
 $ year     : num [1:16] 2008 2009 2010 2011 2012 ...
 $ mean_temp: num [1:16] -1.508 -0.639 2.638 -1.771 -0.461 ...
```

```
tibble [15 x 3] (S3: tbl_df/tbl/data.frame)
 $ year: num [1:15] 2008 2009 2010 2011 2012 ...
 $ mean: num [1:15] -0.656 -0.564 3.395 -1.617 0.518 ...
 $ sd  : num [1:15] 9.29 7.99 6.67 8.01 8.96 ...
```

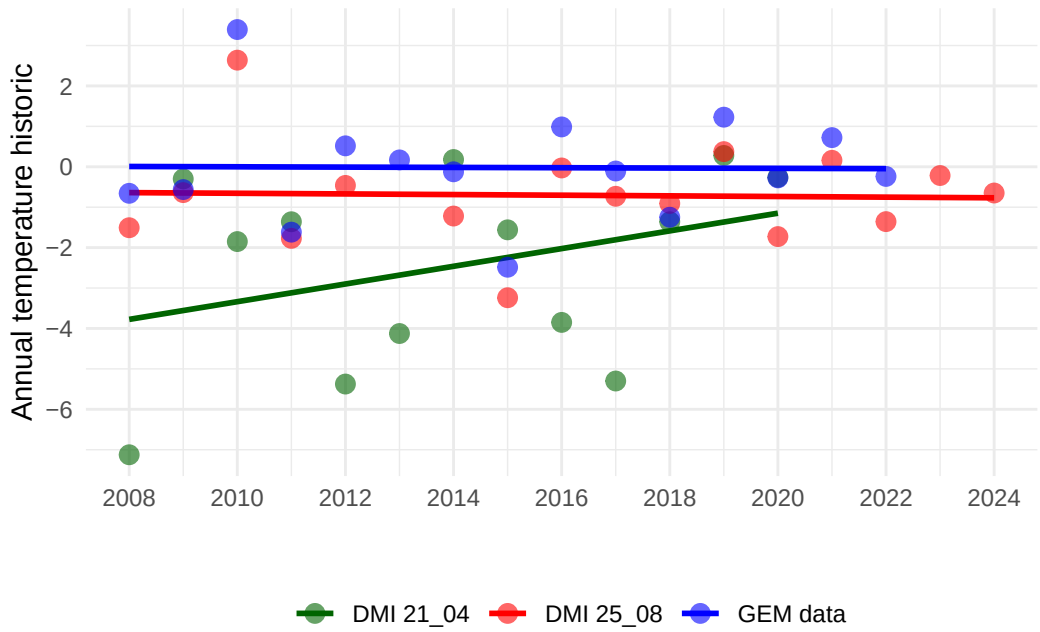

Figure 28: Annual historic temperature in °C only from 2008 to current. DMI 25-08 (Nuuk) and GEM data (Kangerluarsunguaq) corresponds very well with only a slight off set.

7 TO DOS

[ ]make all plots similar. Ref. mnds plots from NERO analysis.

[ ]

download new PAR dataset.

[ ]

make all the heat map the same axis.

References

Cappelen, John. 2021. "Greenland – DMI Historical Climate Data Collection 1784-2020." 21–04. Danish Meteorological Institute. https://www.dmi.dk/fileadmin/Rapporter/2021/DMIRep21-04.pdf.

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